Databases

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DynamoDB – in short



- NoSQL database, fully managed, massive scale (1,000,000 rps)
- Similar to Apache Cassandra (can migrate to DynamoDB)
- No disk space to provision, max object size is 400 KB
- Capacity: provisioned (WCU, RCU, & Auto Scaling) or on-demand
- Supports CRUD (Create Read Update Delete)
- Read: eventually or strong consistency
- Supports transactions across multiple tables (ACID support)
- Backups available, point in time recovery
- Table classes: Standard and Infrequent Access (IA)

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DynamoDB - Basics

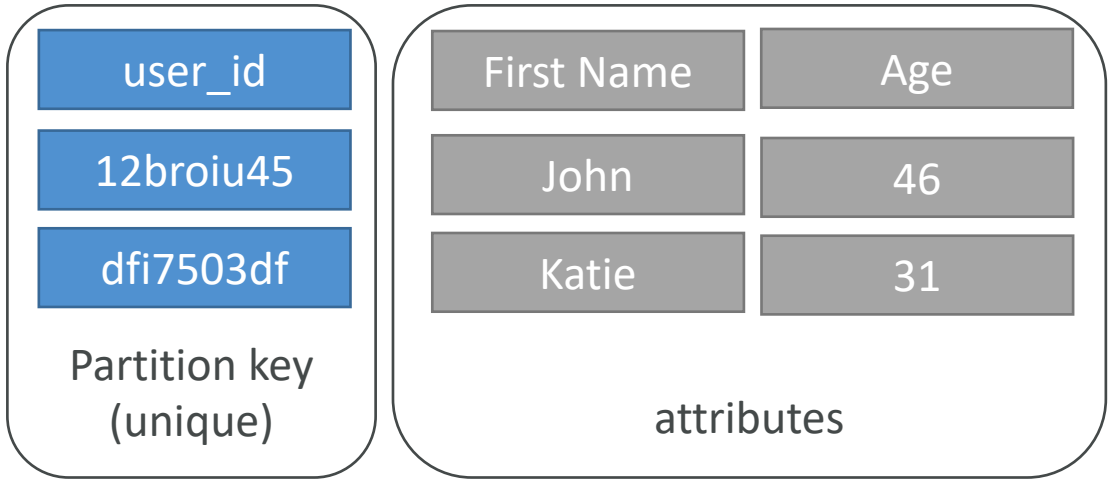
- DynamoDB is made of **tables**
- Each table has a **Primary Key** (must be decided at creation time)
- Each table can have an infinite number of items (= rows)
- Each item has **attributes** (can be added over time – can be null)
- Maximum size of a item is 400KB
- Data types supported are:
  - Scalar Types: String, Number, Binary, Boolean, Null
  - Document Types: List, Map
  - Set Types: String Set, Number Set, Binary Set

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DynamoDB – Primary Keys

- **Option 1: Partition Key only (HASH)**
- Partition key must be unique for each item
- Partition key must be "diverse" so that the data is distributed
- Example: user_id for a users table

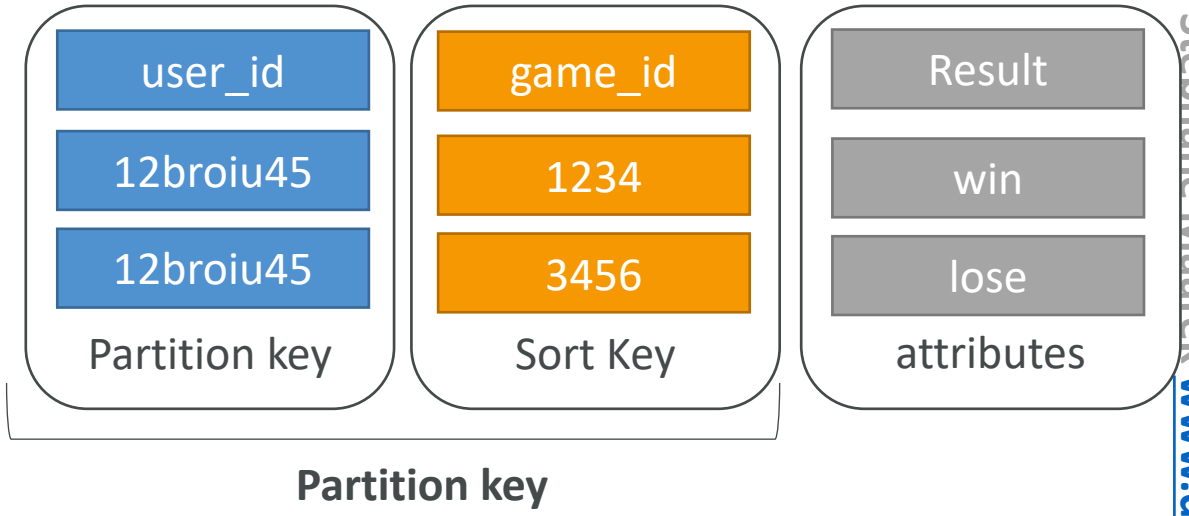


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DynamoDB – Primary Keys

- Option 2: Partition Key + Sort Key
- The combination must be unique
- Data is grouped by partition key
- Sort key == range key
- Example: users-games table
  - user_id for the partition key
  - game_id for the sort key



- Example good sort key: timestamp

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DynamoDB – Indexes

- Object = partition key + optional sort key + attributes
- LSI – Local Secondary Index
  - Keep the same partition key
  - Select an alternative sort key
  - Must be defined at table creation time
- GSI – Global Secondary Index
  - Change the partition key and optionally the sort key
  - Can be defined after the table is created

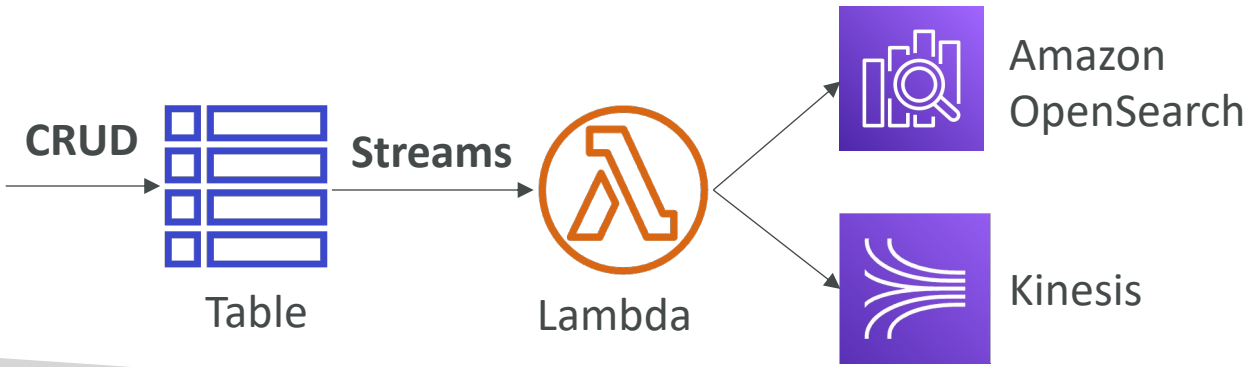
- You can only query by PK + sort key on the main table & indexes **(≠ RDS)**

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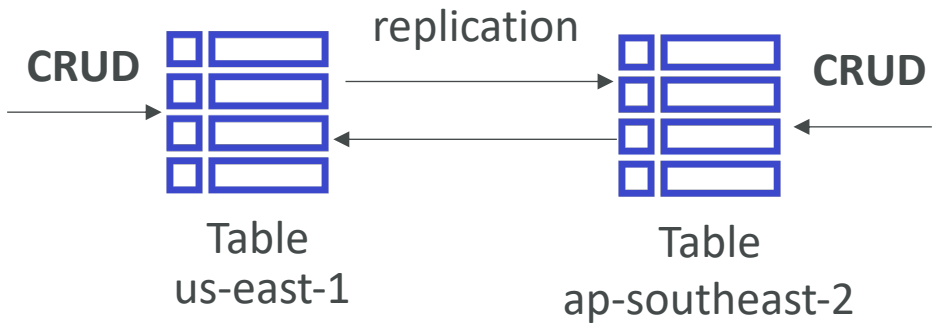
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DynamoDB – Important Features

- **TTL:** automatically expire row after a specified epoch date
- **DynamoDB Streams:**
  - react to changes to DynamoDB tables in real time
  - Can be read by AWS Lambda, EC2…
  - 24 hours retention of data



- **Global Tables:** (cross region replication)
  - Active Active replication, many regions
  - Must enable DynamoDB Streams
  - Useful for low latency, DR purposes

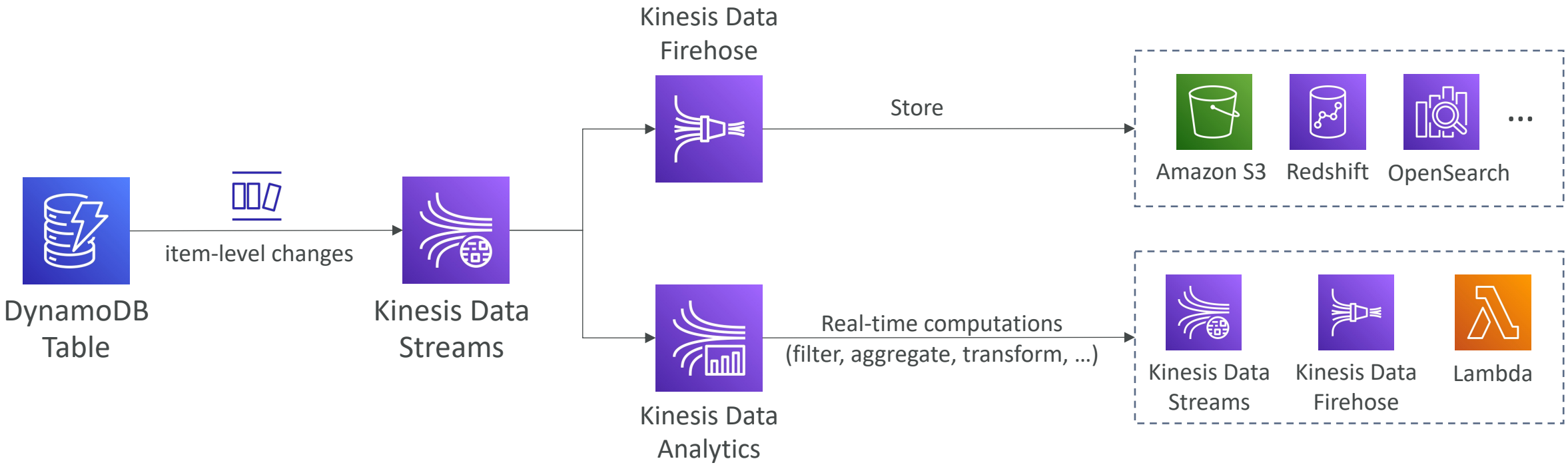


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Amazon Kinesis Data Streams for DynamoDB

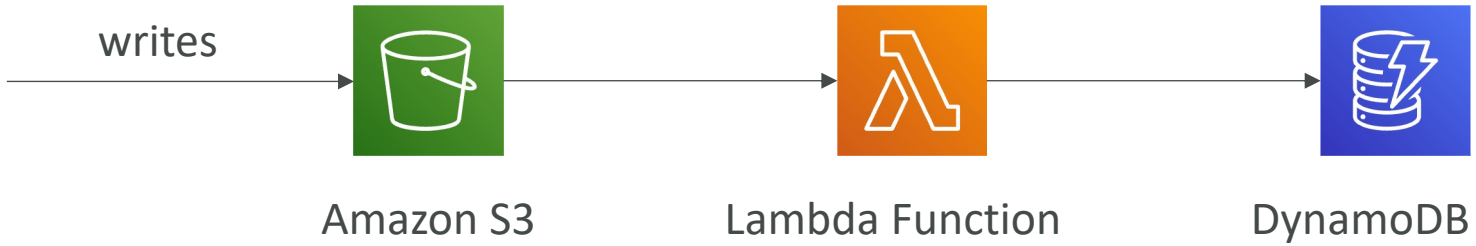
- You can use Kinesis Data Streams to capture item-level changes in DynamoDB
- Custom and longer data retention period (> 24 hours in DynamoDB Streams)



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DynamoDB Solution Architecture
Indexing objects in DynamoDB



API for object metadata

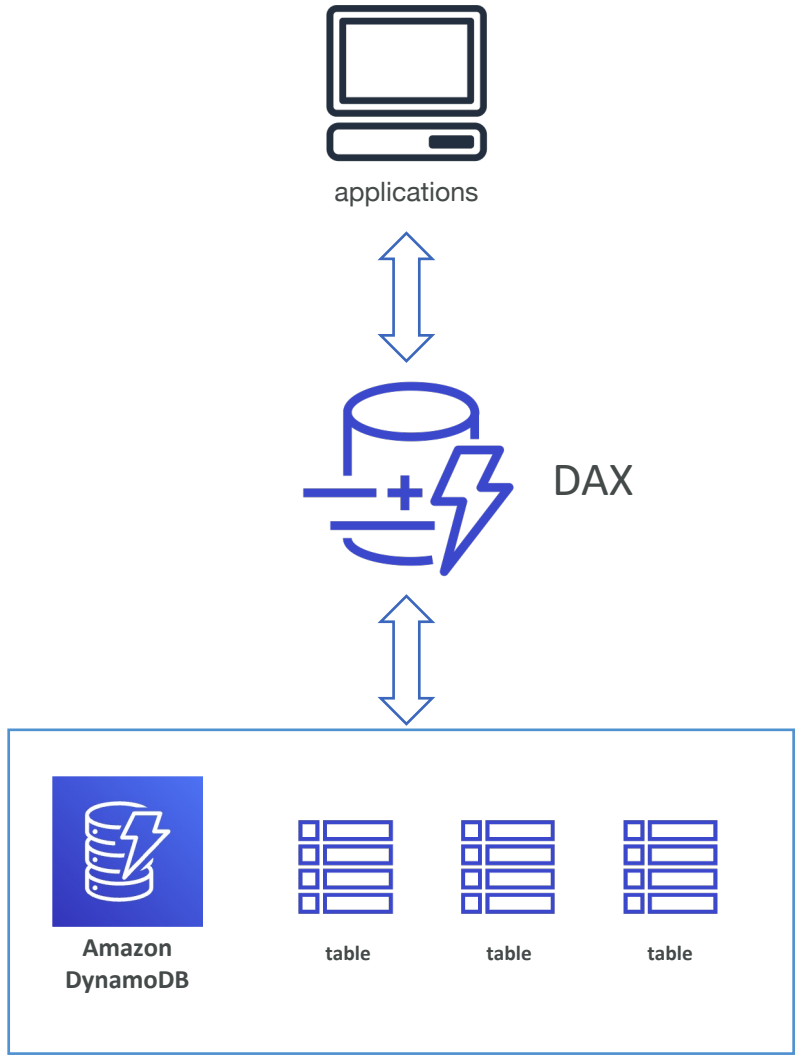
- Search by date
- Total storage used by a customer
- List of all objects with certain attributes
- Find all objects uploaded within a date range

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DynamoDB - DAX

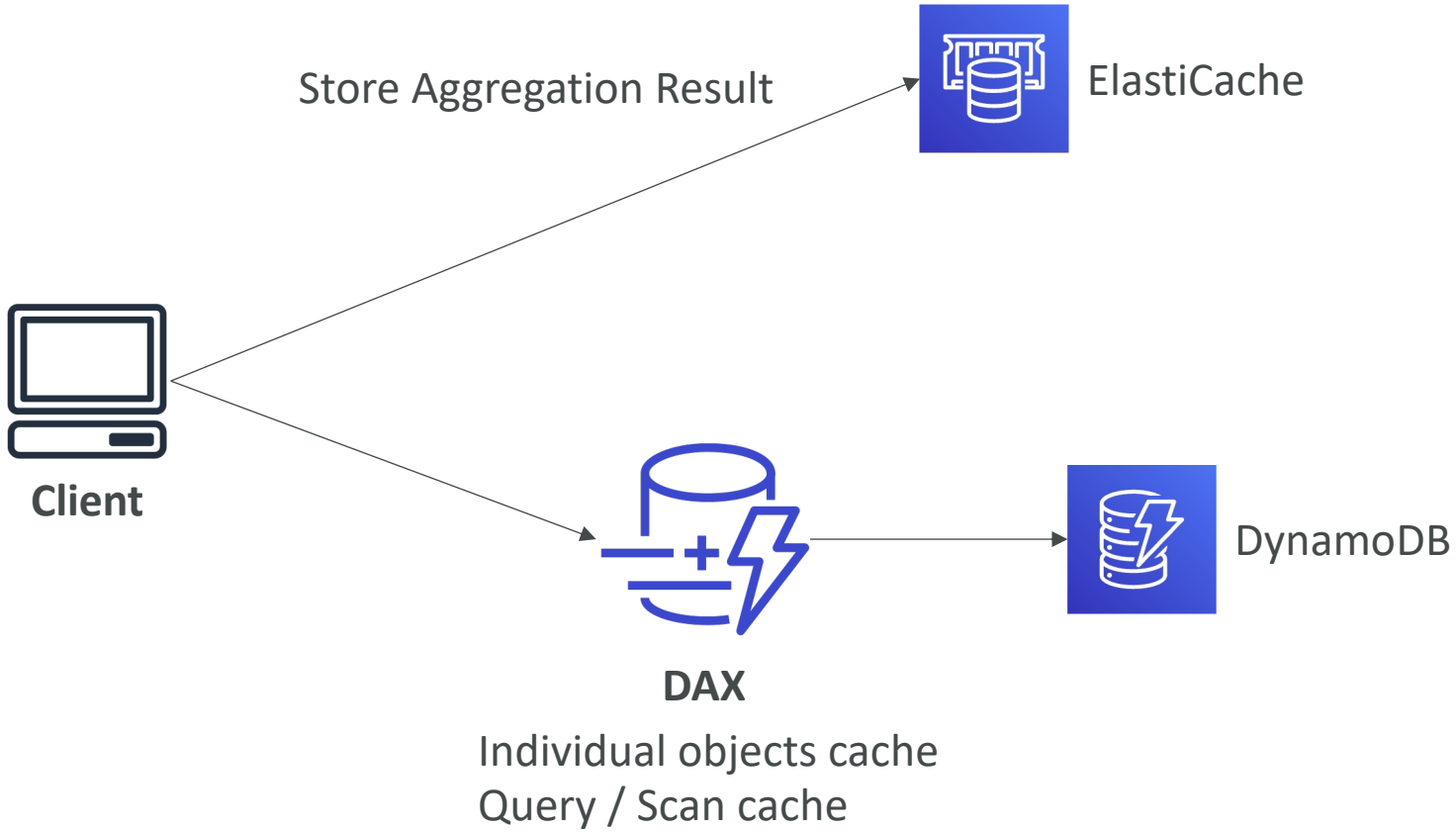
- DAX = DynamoDB Accelerator
- Seamless cache for DynamoDB, no application re-write
- Writes go through DAX to DynamoDB
- Micro second latency for cached reads & queries
- Solves the Hot Key problem (too many reads)
- 5 minutes TTL for cache by default
- Up to 10 nodes in the cluster
- Multi AZ (3 nodes minimum recommended for production)
- Secure (Encryption at rest with KMS, VPC, IAM, CloudTrail…)



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DynamoDB – DAX vs ElastiCache



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Amazon OpenSearch (ex ElasticSearch)



- New name is Amazon OpenSearch
- ElasticSearch => OpenSearch
- Kibana => OpenSearch Dashboards

- Managed version of OpenSearch (open-source project, fork of ElasticSearch)
- Two modes: **Managed** cluster or **Serverless** cluster
- Use cases:
  - Log Analytics
  - Real Time application monitoring
  - Security Analytics
  - Full Text Search
  - Clickstream Analytics
  - Indexing

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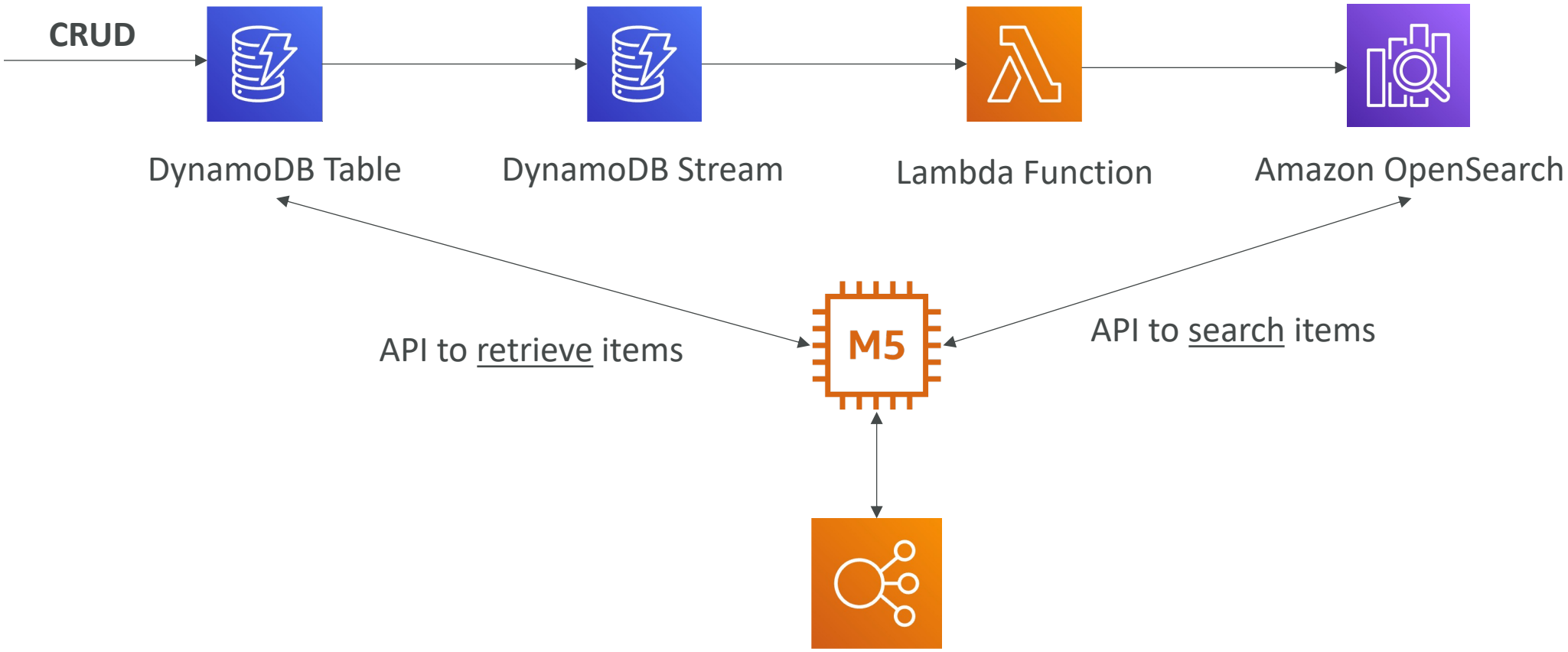
OpenSearch + OS Dashboards + Logstash

- OpenSearch (ex ElasticSearch): provide search and indexing capability
- OpenSearch Dashboards (ex Kibana):
  - Provide real-time dashboards on top of the data that sits in OpenSearch
  - Alternative to CloudWatch dashboards (more advanced capabilities)
- Logstash:
  - Log ingestion mechanism, use the "Logstash Agent"
  - Alternative to CloudWatch Logs (you decide on retention and granularity)

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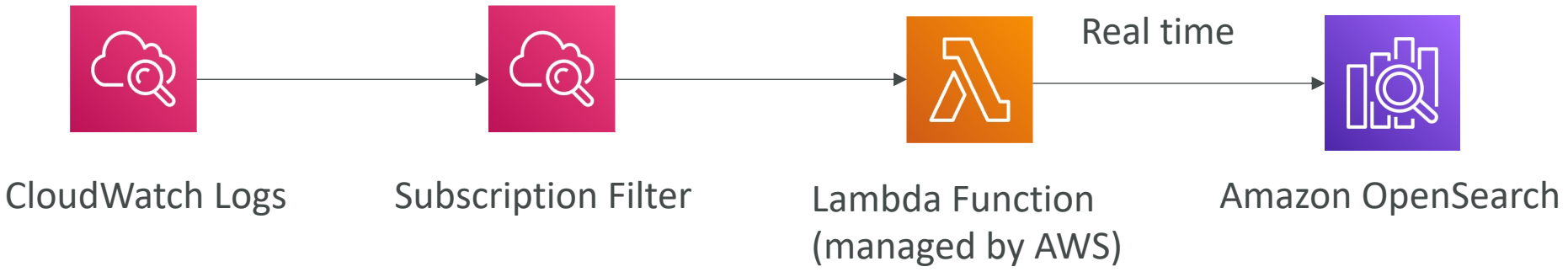
OpenSearch patterns DynamoDB



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OpenSearch patterns CloudWatch Logs





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RDS



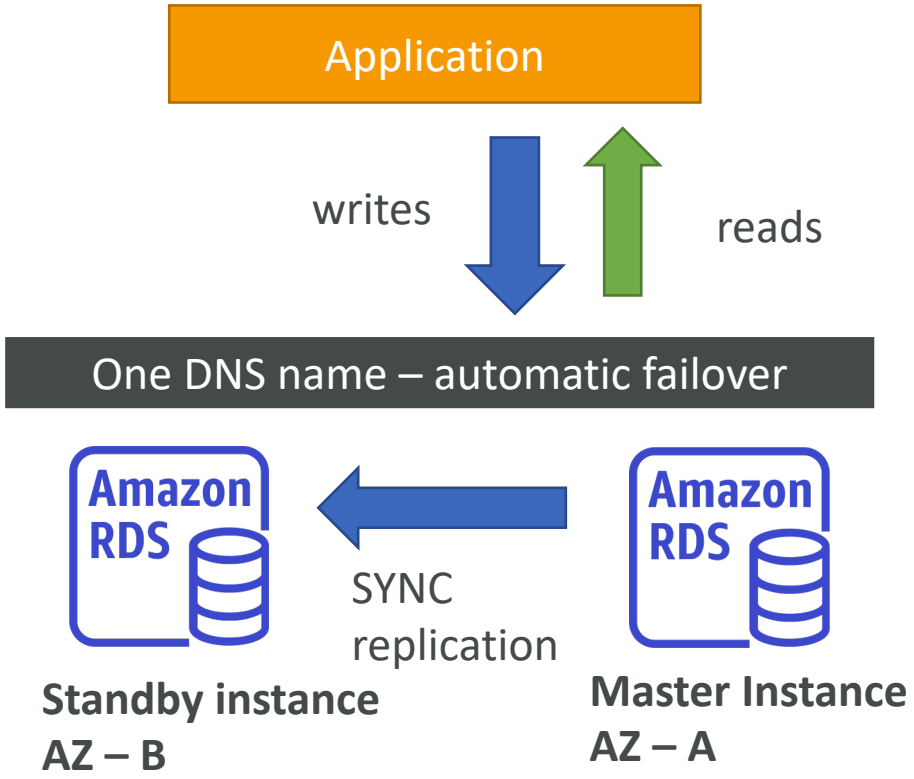
- **Engines:** PostgreSQL, MySQL, MariaDB, IBM DB2, Oracle, SQL Server
- **Managed DB:** provisioning, backups, patching, monitoring
- **Launched within a VPC**, usually in private subnet, control network access using security groups (important when using Lambda)
- **Storage by EBS**, can increase volume size with auto-scaling
- **Backups:** automated with point-in-time recovery. Backups expire
- **Snapshots:** manual, can make copies of snapshots cross region
- **RDS Events:** get notified via SNS for events (operations, outages…)

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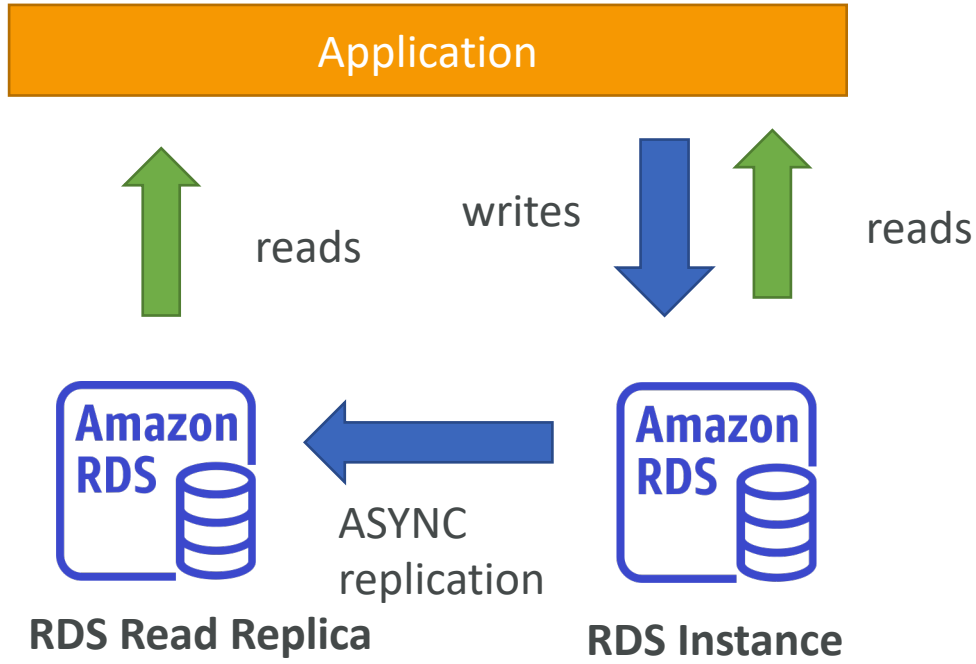
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RDS – Multi AZ & Read Replicas

- **Multi-AZ:** Standby instance for failover in case of outage



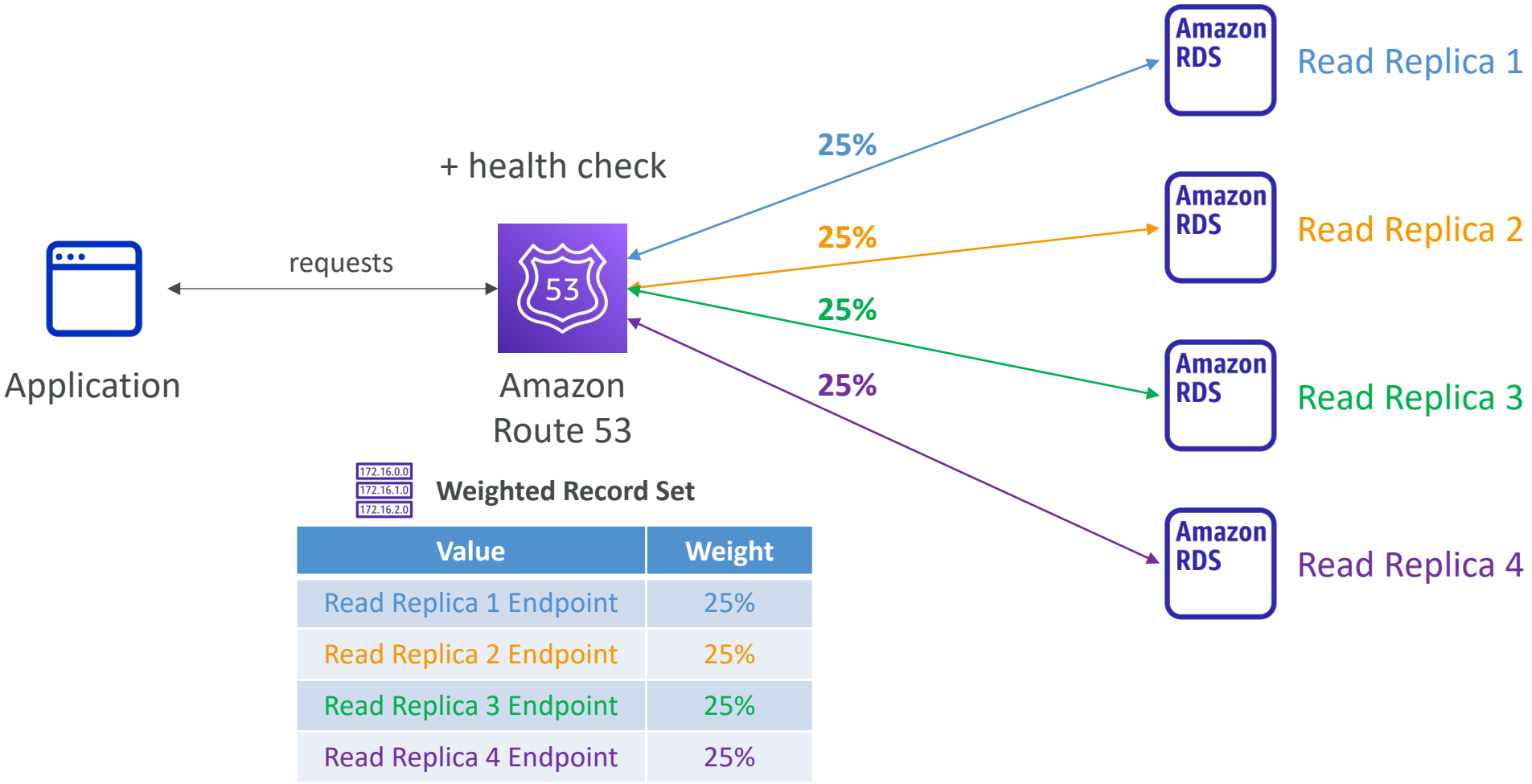
- **Read Replicas:** Increase read throughput. Eventual consistency. Can be cross-region



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RDS – Distributing Reads across Replicas



172.16.0.0
172.16.1.0
172.16.2.0

Weighted Record Set

Value	Weight
Read Replica 1 Endpoint	25%
Read Replica 2 Endpoint	25%
Read Replica 3 Endpoint	25%
Read Replica 4 Endpoint	25%

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RDS – Security (reminder)



- KMS encryption at rest for underlying EBS volumes / snapshots
- Transparent Data Encryption (TDE) for Oracle and SQL Server
- SSL encryption to RDS is possible for all DB (in-flight)
- IAM authentication for MySQL, PostgreSQL and MariaDB
- Authorization still happens within RDS (not in IAM)
- Can copy an un-encrypted RDS snapshot into an encrypted one
- CloudTrail cannot be used to track queries made within RDS

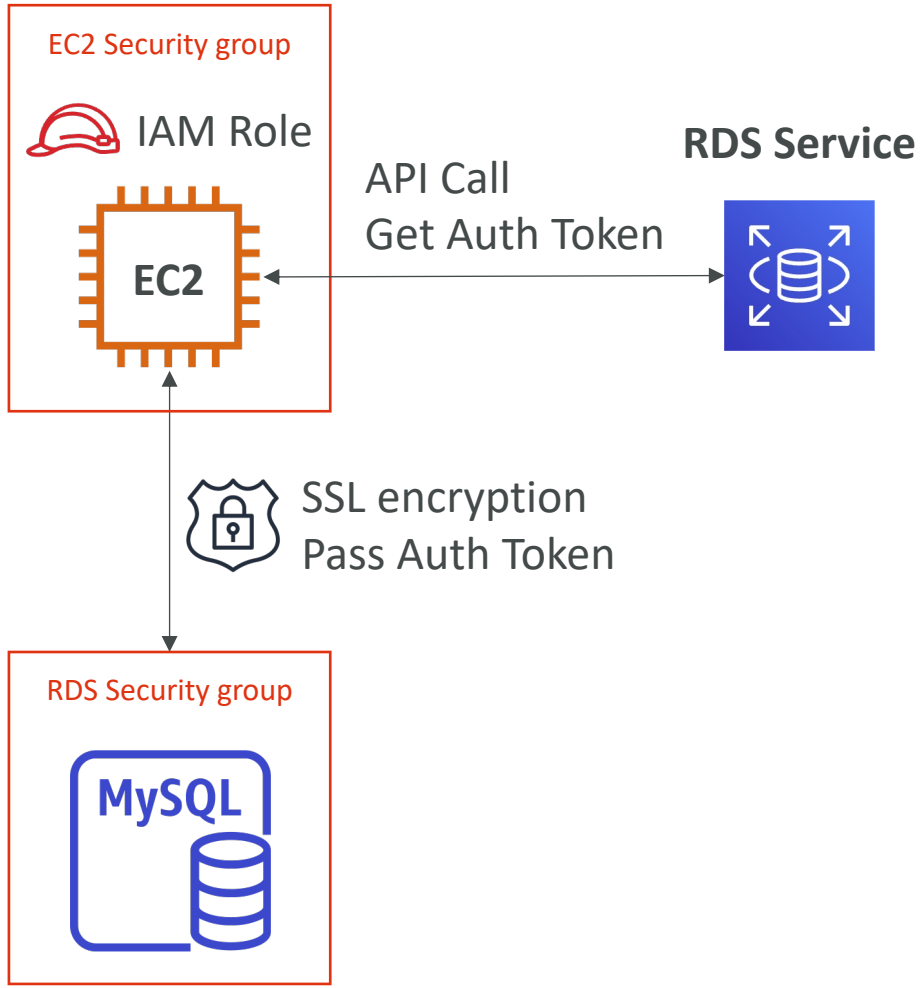
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RDS - IAM Authentication

- IAM database authentication works with **MariaDB, MySQL and PostgreSQL**
- You don't need a password, just an authentication token obtained through IAM & RDS API calls
- Auth token has a lifetime of 15 minutes

- Benefits:
  - Network in/out must be encrypted using SSL
  - IAM to centrally manage users instead of DB
  - Can leverage IAM Roles and EC2 Instance profiles for easy integration



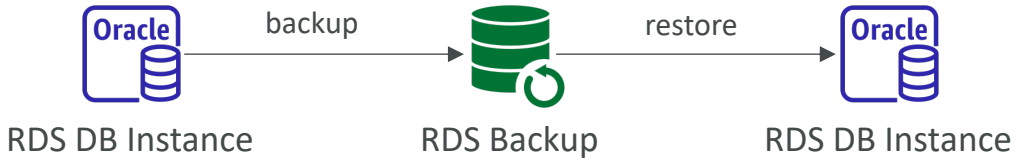
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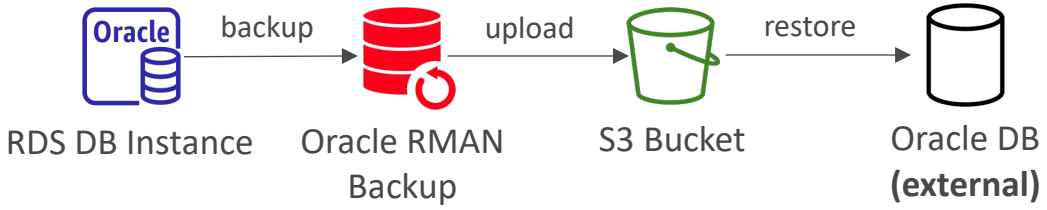
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About RDS for Oracle – Exam Tips

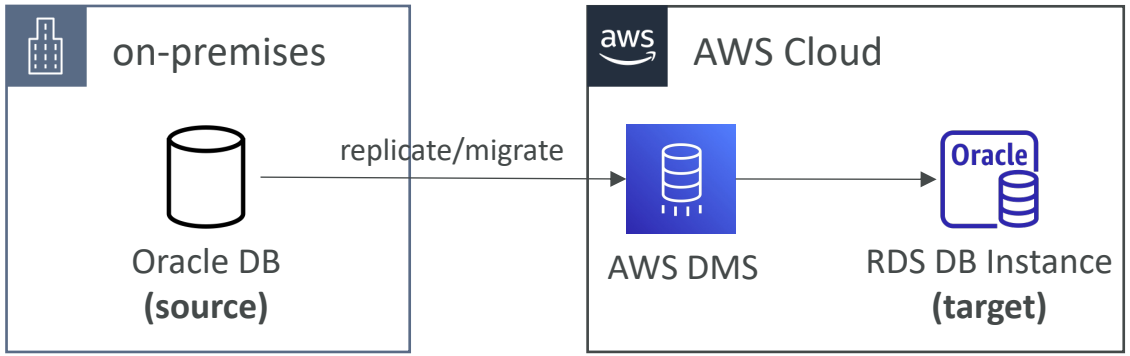
Backups

- Use RDS Backups for backups & restore to Amazon RDS for Oracle
- Use Oracle RMAN (Recovery Manager) for backups & restore to-non RDS (RDS not supported)





- Real Application Clusters (RAC)
  - RDS for Oracle does NOT support RAC
  - RAC is working on Oracle on EC2 Instances because you have full control
- RDS for Oracle supports Transparent Data Encryption (TDE) to encrypt data before it's written to storage
- DMS works on Oracle RDS

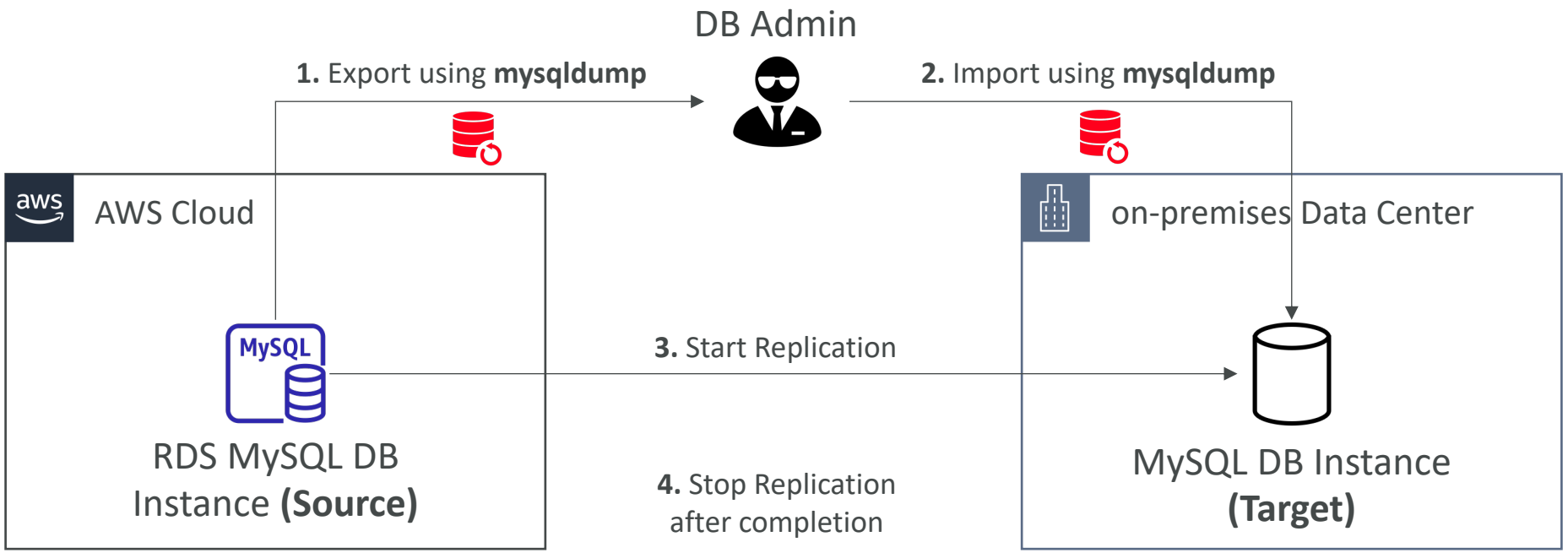


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About RDS for MySQL

- You can use the native **mysqldump** to migrate a MySQL RDS DB to non-RDS
- The external MySQL database can run either on-premises in your data center, or on an Amazon EC2 instance

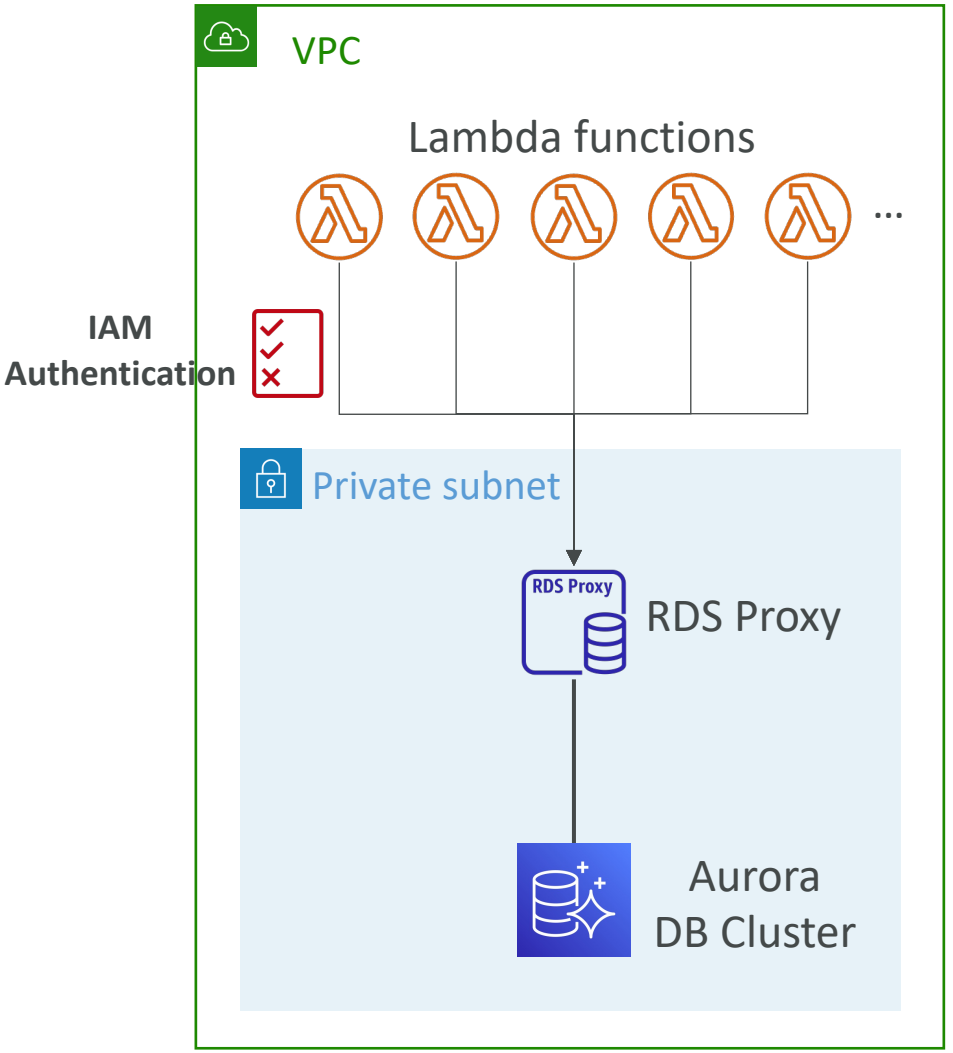


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RDS Proxy for AWS Lambda

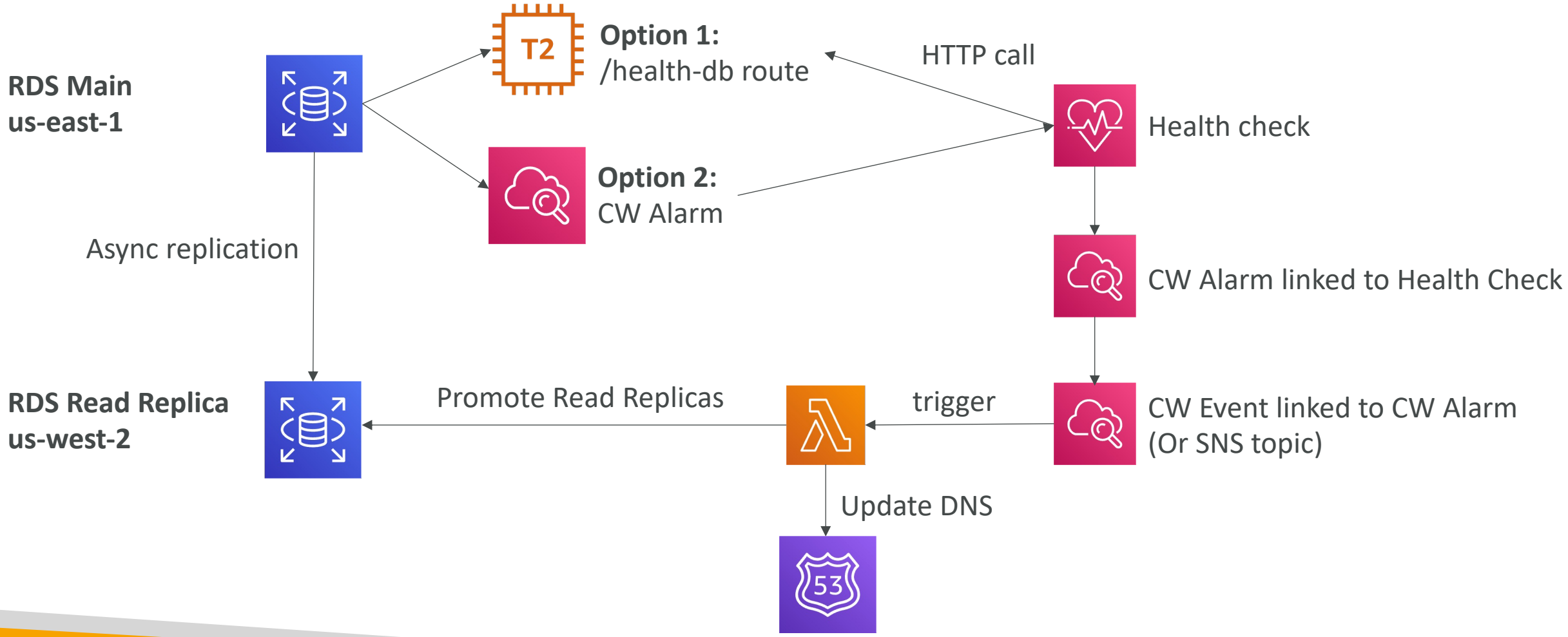
- When using Lambda functions with RDS, it opens and maintains a database connection
- This can result in a "TooManyConnections" exception
- With RDS Proxy, you no longer need code that handles cleaning up idle connections and managing connection pools
- Supports IAM authentication or DB authentication, auto-scaling
- The Lambda function must be deployed in your VPC, because RDS Proxy is never publicly accessible



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RDS Solution Architecture
Cross Region Failover



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Aurora



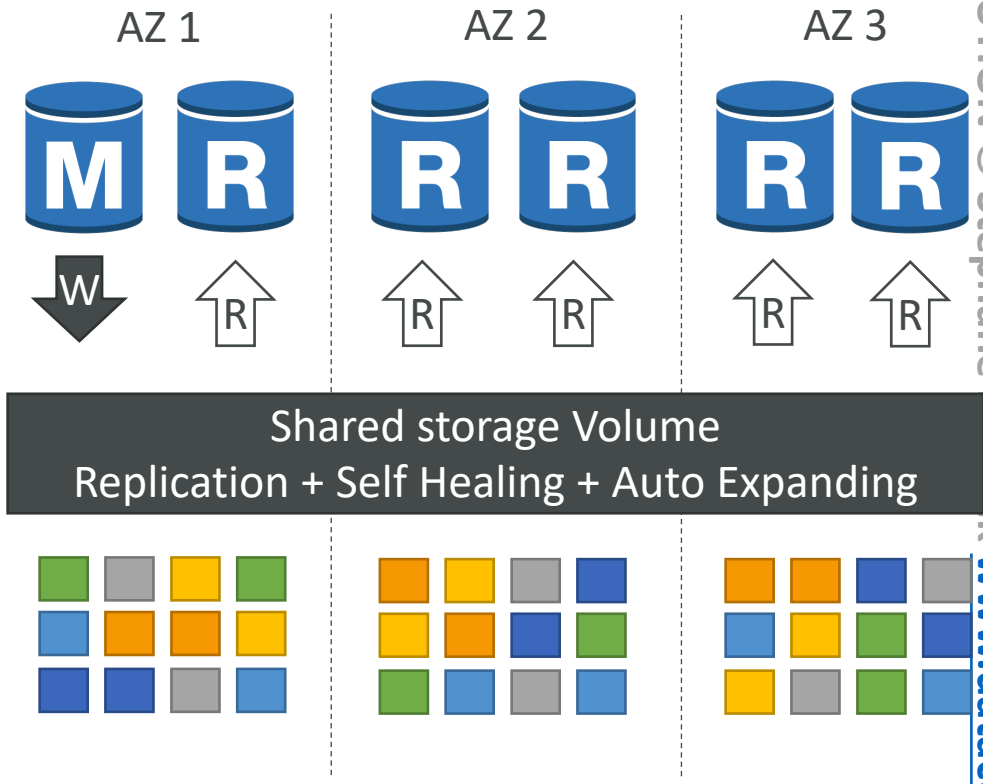
- **DB Engines:** PostgreSQL-compatible & MySQL-compatible
- **Storage:** automatically grows up to 256 TB, 6 copies of data, multi-AZ
- **Read Replicas:** up to 15 RR, reader endpoint to access them all
- **Cross Region RR:** entire database is copied (not select tables)
- **Load / Offload data directly from / to S3:** efficient use of resources
- **Backup, Snapshots & Restore:** same as RDS

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Aurora High Availability and Read Scaling

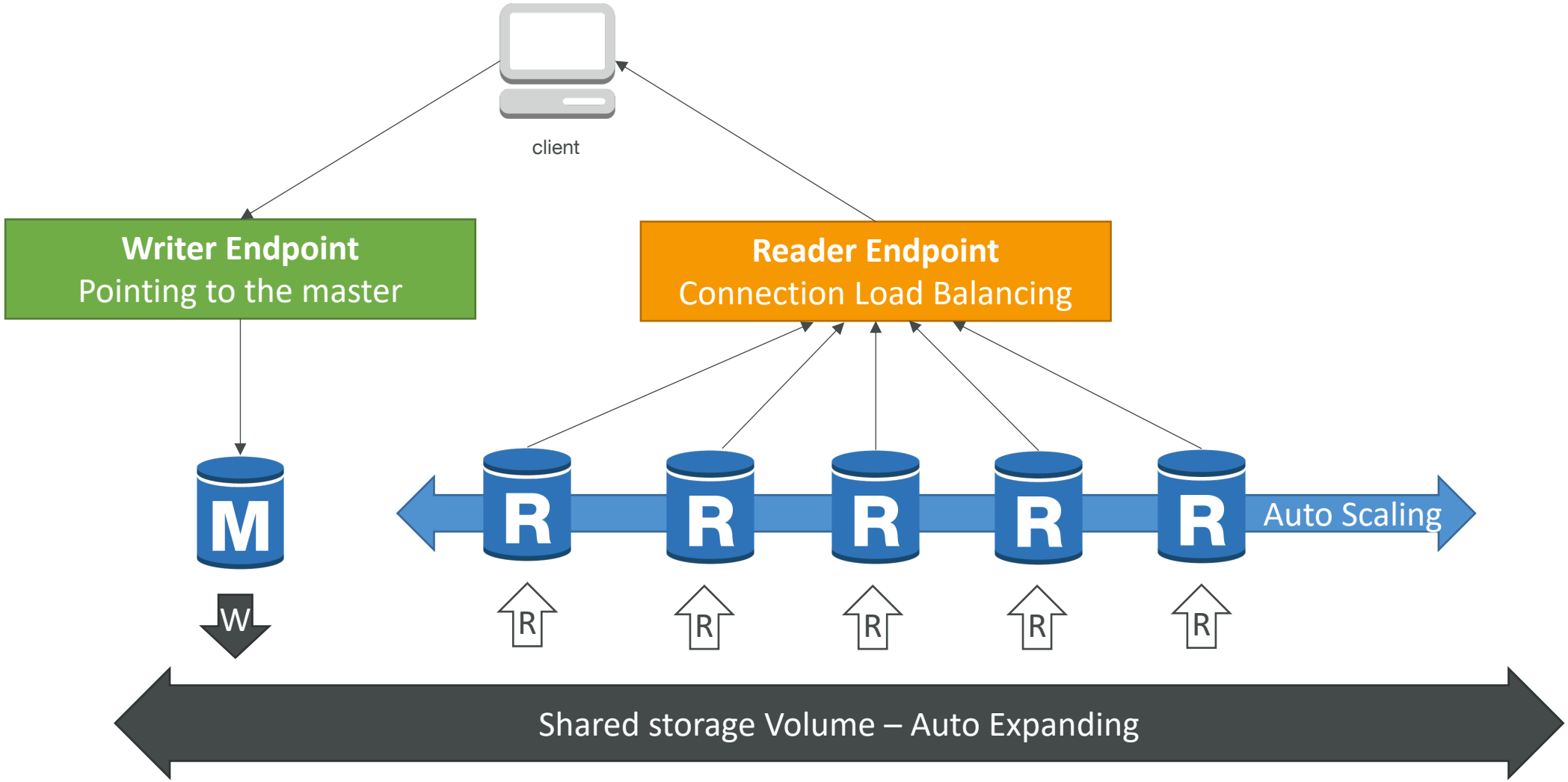
- 6 copies of your data across 3 AZ:
  - 4 copies out of 6 needed for writes
  - 3 copies out of 6 need for reads
  - Self healing with peer-to-peer replication
  - Storage is striped across 100s of volumes
- Automated failover for master in less than 30 seconds
- Master + up to 15 Aurora Read Replicas serve reads
- Support for Cross Region Replication



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Aurora DB Cluster



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Aurora Endpoints

- Endpoint = Host Address + Port

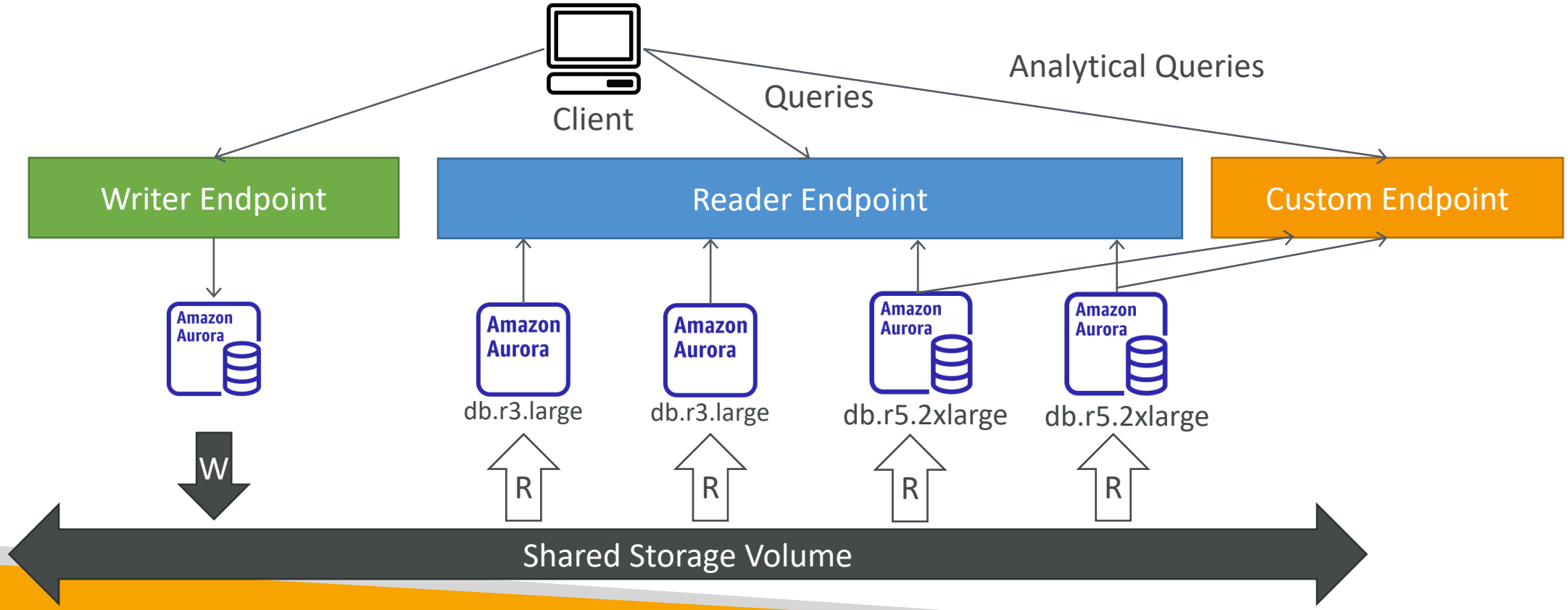
- Cluster Endpoint (Writer Endpoint)
  - Connects to the current primary DB instance in the Aurora cluster
  - Used for all write operations in the DB cluster (inserts, updates, deletes, and queries)
- Reader Endpoint
  - Provides load-balancing for read only connections to all Aurora Replicas in the Aurora cluster
  - Used only for read operations (queries)
- Custom Endpoint
  - Represents a set on DB instances that you choose in the Aurora cluster
  - Used when you want to connect to different subsets of DB instances with different capacities and configurations (e.g., different DB parameter group)
- Instance Endpoint
  - Connects to a specific DB instance in the Aurora cluster
  - Used when you want to diagnosis and fine tune a specific DB instance

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Aurora – Custom Endpoints

- Define a subset of Aurora Instances as a Custom Endpoint
- Example: Run analytical queries on specific replicas
- The Reader Endpoint is generally not used after defining Custom Endpoints



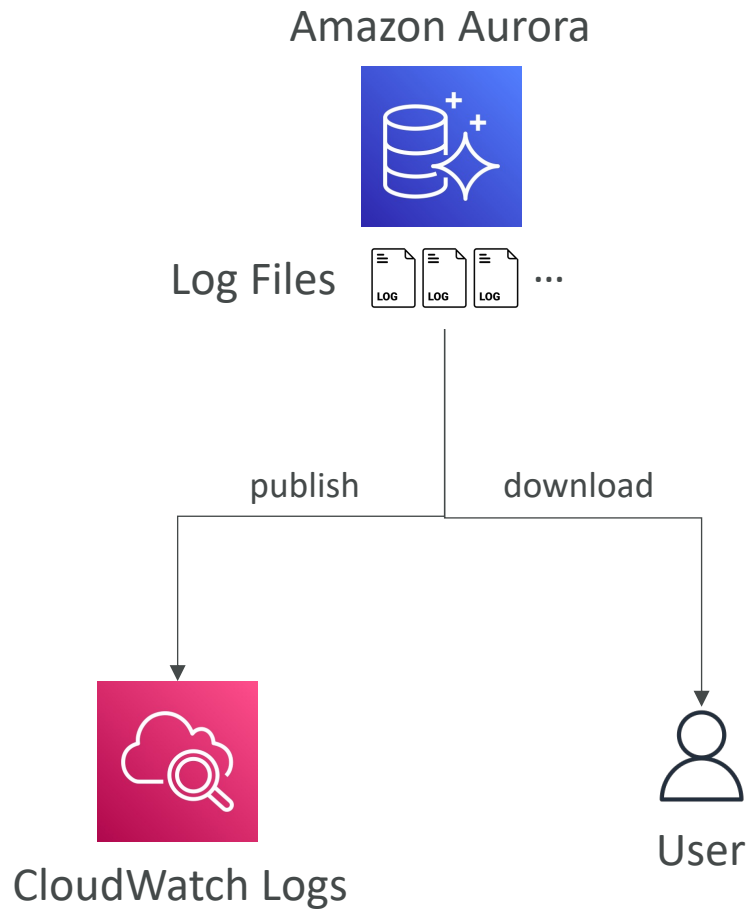
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Aurora Logs

- You can monitor the following types of Aurora MySQL log files:
  - Error log
  - Slow query log
  - General log
  - The audit log

- These log files are either downloaded or published to CloudWatch Logs

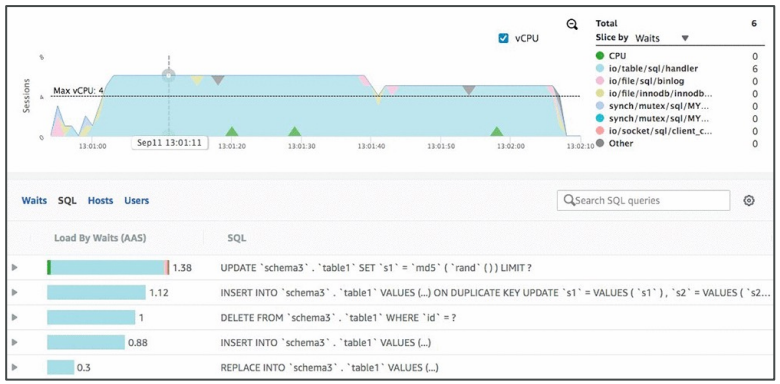


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Troubleshooting RDS & Aurora Performance

- **Performance Insights:** find issues by waits, SQL statements, hosts and users



SQL Statements



Users

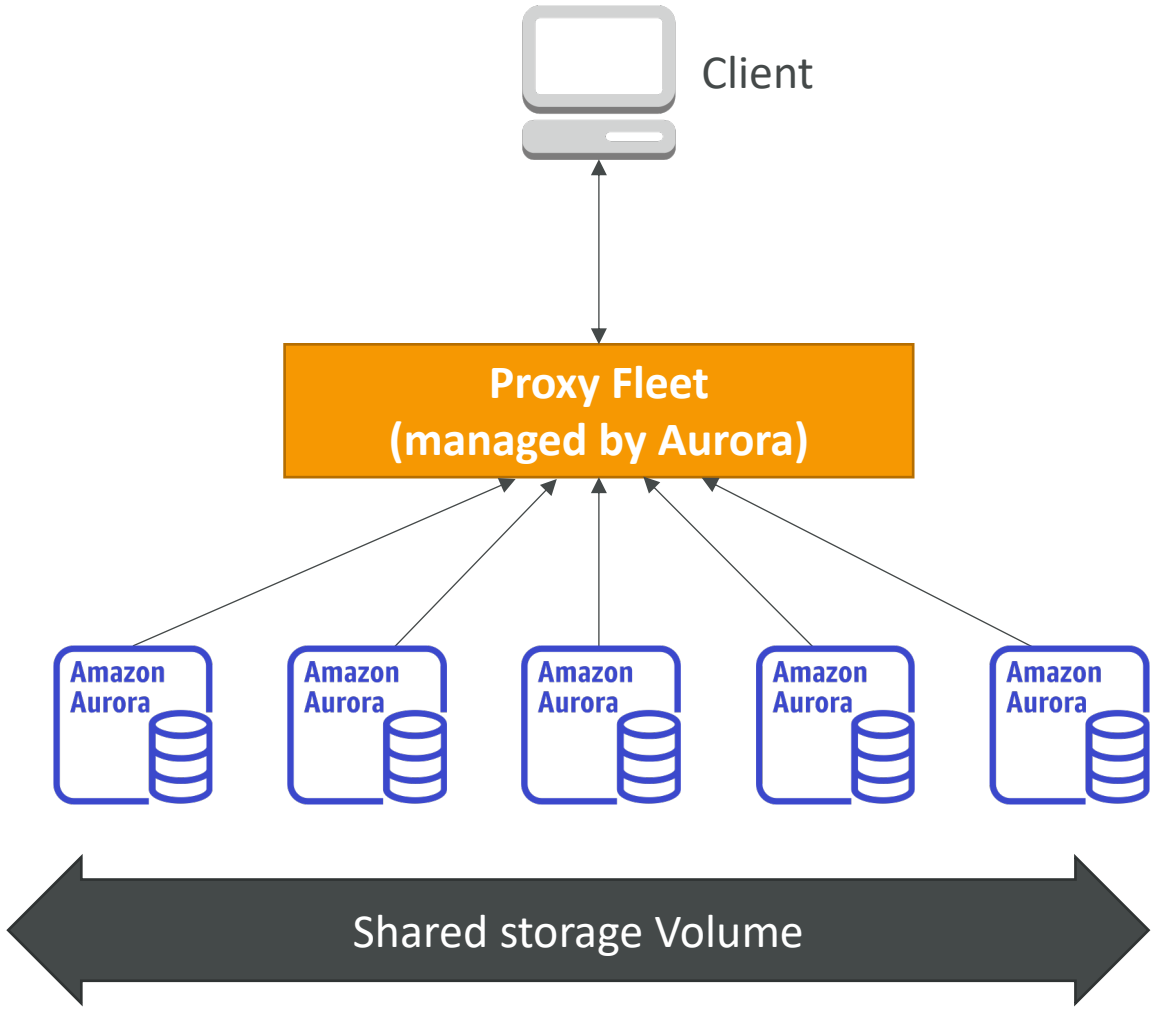
- **CloudWatch Metrics:** CPU, Memory, Swap Usage
- **Enhanced Monitoring Metrics:** at host level, process view, per-second metric
- Slow Query logs

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Aurora Serverless

- Automated database instantiation and auto-scaling based on actual usage
- Good for infrequent, intermittent or unpredictable workloads
- No capacity planning needed
- Pay per second, can be more cost-effective

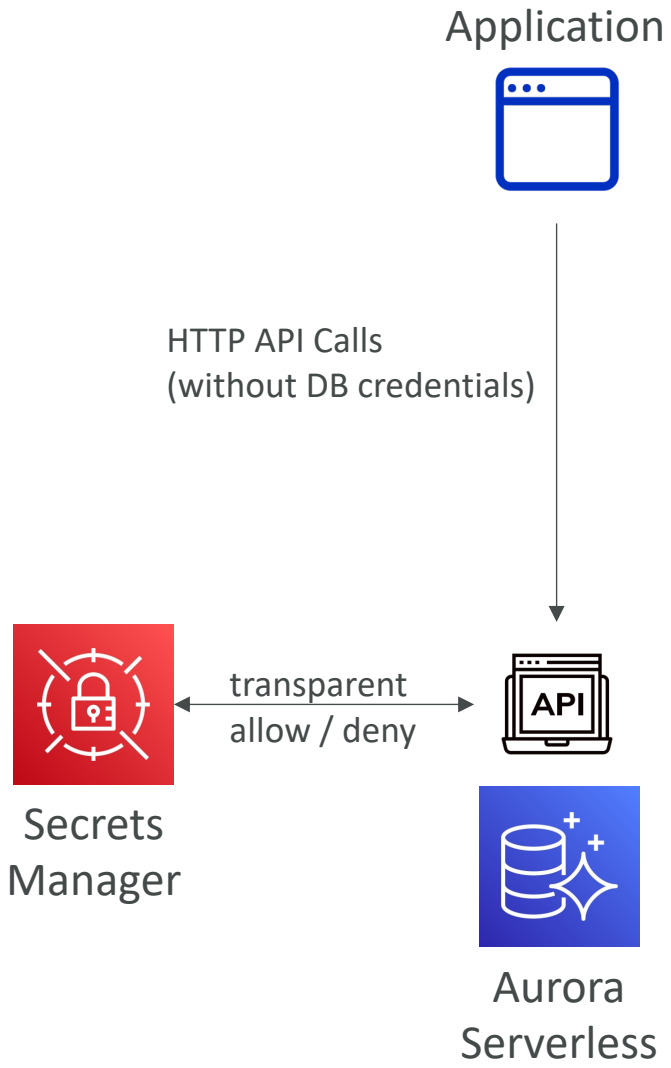


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Aurora Serverless – Data API

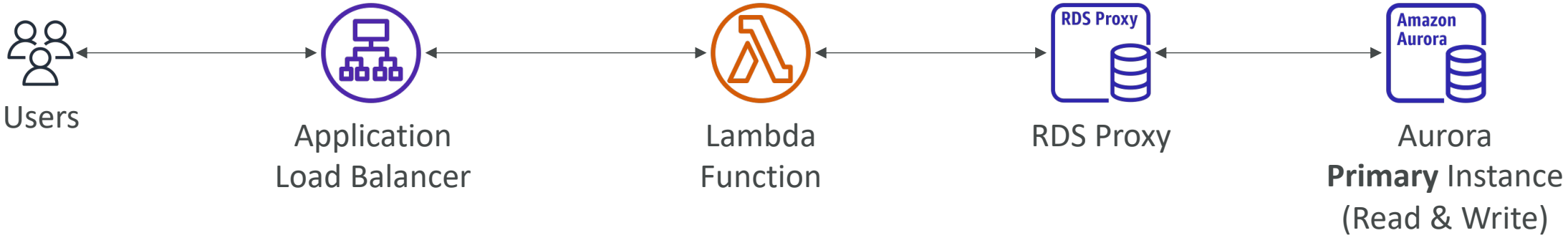
- Access Aurora Serverless DB with a simple API endpoint (no JDBC connection needed)
- **Secure HTTPS endpoint to run SQL statements**
- No persistent DB connections management
- Users must be granted permissions to Data API and Secrets Manager (where credentials are checked)



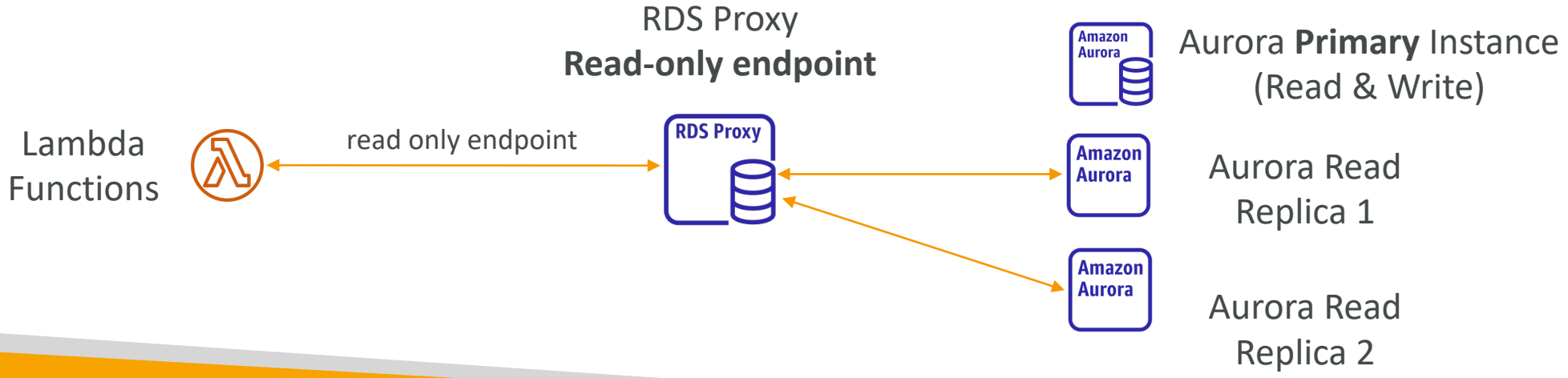
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RDS Proxy for Aurora



- Ability to create an additional read-only endpoint that connects to Aurora Read Replicas only

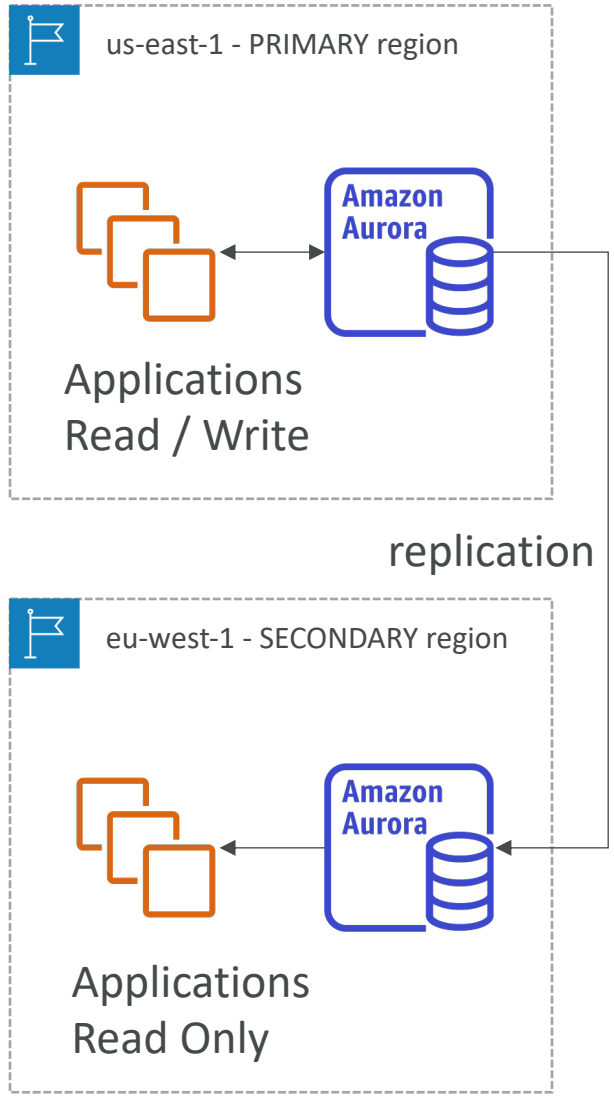


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Global Aurora

- Aurora Cross Region Read Replicas
  - Useful for disaster recovery
  - Simple to put in place
- Aurora Global Database (recommended)
  - 1 Primary Region (read / write)
  - Up to 10 secondary (read-only) regions, replication lag is less than 1 second
  - Up to 16 Read Replicas per secondary region
  - Helps for decreasing latency
  - Promoting another region (for disaster recovery) has an RTO of < 1 minute
  - Ability to manage the RPO in Aurora for PostgreSQL

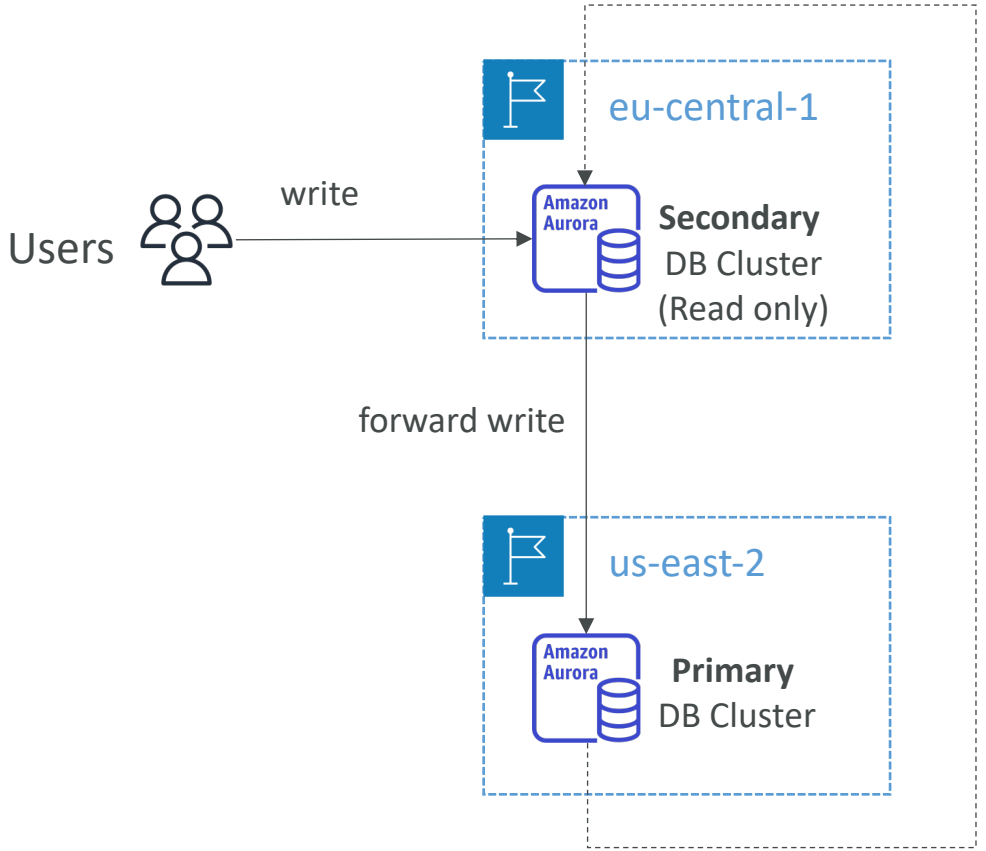


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Aurora Global – Write Forwarding

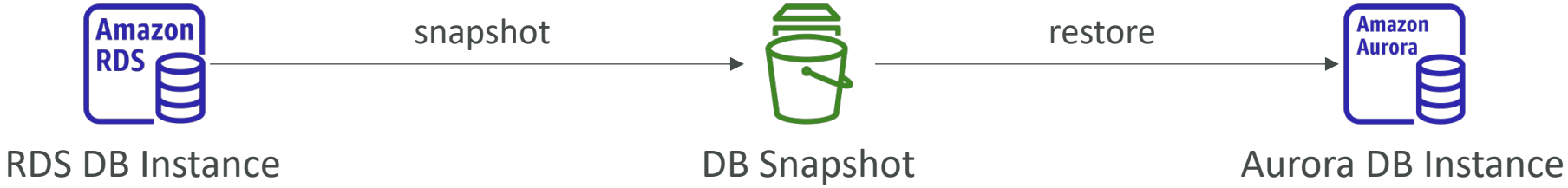
- Enables Secondary DB Clusters to forward SQL statements that perform write operations to the Primary DB Cluster
- Data is always changed first on the Primary DB Cluster, then replicated to the Secondary DB Clusters
- Primary DB Cluster always has an up-to-date copy of all data
- Reduces the number of endpoints to manage



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Convert RDS to Aurora





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